

Product Requirements Document (PRD)

Personal Portfolio — Jassim M. Shamim

Digital Accessibility Engineer & Creative Technologist

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Inspiration	williamle.design (spatial 3D entry + Playground), juanmora.co (scroll-driven Work)

1. Vision & Positioning

A portfolio that proves the thesis of my work: that beautiful, cutting-edge digital experiences and universal accessibility are the same goal, not a trade-off. It fuses two inspirations — William Le’s spatial, Apple-style 3D entry and masonry “Playground,” and Juan Mora’s cinematic scroll-driven “Work” page with a sticky project index — while being fully keyboard-operable, screen-reader-friendly, reduced-motion-safe, and WCAG 2.2 AA compliant.

Strategic differentiator: Both reference sites are visually outstanding but accessibility-thin (verified: neither declares a `lang` attribute; Juan’s site uses almost no landmark semantics and ships no skip link; both lean on autoplaying motion). Because I build accessibility for a living, **the portfolio itself is my strongest proof of skill**. It will match their craft AND be exemplary on accessibility. That contrast is the story.

Tagline concept: *“Building a digital world that’s beautiful and built for everyone.”*

2. Goals & Success Metrics

Primary goal: Recruiter / hiring-manager inbound and credibility for accessibility-engineering and creative-technologist roles.

Secondary goal: A living home for experiments, design curation, and side projects.

Success metrics:

- Lighthouse Accessibility = 100; Performance >= 90 (mobile).
 - Zero automated axe-core violations in CI.
 - Full keyboard operability, no focus traps.
 - Core Web Vitals green: LCP < 2.5s, CLS < 0.1, INP < 200ms.
 - Qualitative: a recruiter can grasp who I am and see three projects within 60 seconds.
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3. Personas

1. **Hiring manager / recruiter (primary):** desktop, time-pressed, needs proof of engineering depth + design taste fast.
 2. **Fellow engineer / designer:** will inspect the craft, possibly via keyboard or screen reader.
 3. **Me (maintainer):** needs a low-friction way to add projects and playground items over time.
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4. Information Architecture / Sitemap

Flat, anchor-driven structure (mirrors both references) to keep complexity low:

Route	Purpose	Inspiration
/	Spatial 3D hero entry + selected-work teaser + bookend CTA	William Le
/work	Full project showcase with sticky left scroll-spy	Juan Mora
/playground	Masonry grid of experiments, designs, curation	William Le
/about	Story, accessibility philosophy, skills, contact	—

Shared persistent pill-style top nav and a shared bookend footer (CTA + wordmark) appear on every route, so the site feels continuous regardless of entry point.

Decision: Flat single-page-with-anchors per route (no separate case-study pages); projects link out to live demos/repos. Architecture supports adding `/work/[slug]` later without rework.

5. Signature Features (functional + a11y spec)

5.1 — 3D Spatial Hero (inspired by William Le)

A tilted “creative technologist’s desk” diorama: floating panels for an AI/agent terminal, a Figma frame, an accessibility audit/contrast checker, and a code editor, arranged on a plane rotated ~20deg on X with layered depth + subtle mouse-parallax.

- **Implementation:** CSS 3D transforms, NOT WebGL. (Verified: William’s scene uses per-element `matrix3d` encoding `rotateX(20deg) + scale(0.5)` with baked perspective and `will-change: transform ; zero <canvas> .`) Keeps bundle + complexity low.
- **Behavior:** scene assembles on load; “camera” eases inward on scroll toward the work teaser.
- **A11y:** scene is decorative -> `aria-hidden="true"` ; real `<h1>` + tagline as text. Under `prefers-reduced-motion: reduce` , parallax/assembly disabled and a static composed image shown. No info conveyed by animation alone.

5.2 — Work Showcase (inspired by Juan Mora)

Two-column layout: a sticky left index (`position: sticky; top: 0`) + scrolling content. Each project block: Title, Year pill, “View live / repo” button, then **Challenge / Services (tags) / Role**, plus media (looping muted video or images).

- **Behavior:** active project highlights in the index as you scroll (verified pattern on Juan’s site).
- **A11y implementation:** index is a real `<nav aria-label="Projects">` ; active state driven by IntersectionObserver, also sets `aria-current="true"` ; clicking smooth-scrolls (Lenis) and moves focus to the section heading. Each project = `<section aria-labelledby>` with a heading. Videos `muted loop playsinline preload="none"` with poster + text alternative; only play in view and when motion allowed.

5.3 — Playground (inspired by William Le)

3-column masonry (verified: William’s is 3 fixed columns ~495px wide) mixing design experiments, Figma concepts, AI-tool demos, and curation. Each card: preview

(image/muted video), title, medium/tool, date tag; newest first.

- **Implementation:** CSS `columns: 3; gap` with `break-inside: avoid` (zero-JS, degrades to 2 -> 1 column).
- **A11y:** semantic list; each card is one focusable link with a clear accessible name; videos lazy-play in view, pause off-view; visible focus rings; keyboard-only operable.

5.4 — Bookend Footer

A single repeated CTA card (“Let’s build something inclusive together”) + large wordmark/monogram footer on every route (Juan resolves Home and Work to the identical closing moment). Contains `mailto:`, LinkedIn, GitHub, optional contact form. No downloads or sharing flows.

5.5 — Global Chrome

Persistent pill-style top nav (Home / Work / Playground / About) with a left monogram; optional live clock (William detail). Custom cursor is optional + progressive: never replaces the real cursor for keyboard/touch; disabled under reduced-motion.

6. Design Tokens

Built from a **blue / black / beige** identity, blending Juan’s warm beige light surface (sampled `#FAF6EF`) with William’s near-black canvas (`#0E0E0E`). **Dark is the primary mode** (makes 3D panels + work media pop, like William’s); a beige-forward light mode is available. Both AA-compliant. Starting values — tune in design.

```
/* ---- Color (verify each final pairing against WCAG AA) ---- */
--color-ink:    #0E0E0E; /* near-black, primary dark bg (William) */
--color-surface: #14161A; /* raised dark surface / cards */
--color-beige:   #FAF6EF; /* warm beige, light bg (Juan) */
--color-beige-2: #EFE9DC; /* beige raised surface */
--color-blue:    #2D6BFF; /* primary brand / action */
--color-blue-600: #1E4FD0; /* hover / pressed */
--color-blue-200: #AFC6FF; /* accents on dark */
--color-text-hi:  #F5F3EE; /* high-emphasis text on dark */
--color-text-lo:  #A7A39B; /* muted text (large/secondary only) */
--color-text-ink: #1A1A1A; /* text on beige */
--color-focus:    #7FA8FF; /* 3px focus ring, high contrast both modes */
```

```

/* ---- Type ---- */
--font-sans: "Inter", system-ui, sans-serif; /* UI/body */
--font-display: "Clash Display", "Inter", sans-serif; /* headlines */
--font-mono: "JetBrains Mono", monospace; /* code accents */
--fs-display: clamp(2.5rem, 6vw, 6rem);
--fs-h1: clamp(2rem, 4vw, 3.5rem);
--fs-h2: clamp(1.5rem, 3vw, 2.25rem);
--fs-body: 1rem; /* 16px min */
--fs-small: 0.875rem;
--lh-tight: 1.05; --lh-body: 1.6; --tracking: 0.01em;

/* ---- Space (8pt) ---- */
--s-1:.25rem; --s-2:.5rem; --s-3:.75rem; --s-4:1rem; --s-6:1.5rem;
--s-8:2rem; --s-12:3rem; --s-16:4rem; --s-24:6rem; --s-32:8rem;

/* ---- Radius / elevation ---- */
--radius-card: 16px; --radius-pill: 999px;
--shadow-1: 0 2px 8px rgb(0 0 0 / .25);
--shadow-glow: 0 0 80px rgb(45 107 255 / .25); /* Juan's top-glow */

/* ---- Motion (all gated by prefers-reduced-motion) ---- */
--ease-out: cubic-bezier(.22,1,.36,1);
--dur-fast:.2s; --dur-base:.5s; --dur-slow:.9s;

```

Contrast notes: body text at `--color-text-hi` on `--color-ink` passes AA; reserve `--color-text-lo` for large/secondary text; `--color-blue` + white text for actions on dark passes. Verify every final pairing with axe DevTools.

7. Tech Stack (low-complexity)

Recommended: Next.js (App Router) + TypeScript + Tailwind CSS + GSAP/ScrollTrigger + Lenis, deployed on Vercel.

Why Next.js over React+FastAPI here: a portfolio is content-heavy and largely static. Next's static generation gives great SEO, fast LCP, and accessible server-rendered HTML for free. FastAPI is a Python backend; a portfolio has no real backend need beyond perhaps a contact-form handler. Standing up a separate FastAPI service adds complexity for little benefit. If a backend is needed later (contact form, an LLM-powered demo), add a small Next serverless route — or introduce a FastAPI microservice only then.

Layer	Choice	Notes
Framework	Next.js (App Router) + TS	SSG, image optimization, routing
Styling	Tailwind + CSS variables (tokens)	tokens map to Tailwind theme
Motion	GSAP + ScrollTrigger + Lenis	matches Juan's verified stack
Content	MDX / typed content files	add a file = add a project
Hosting	Vercel	free, CI-friendly
Optional backend	Next route handler or FastAPI	only if/when needed

No WebGL, no UnicornStudio required. (If the cinematic pre-rendered hero look is ever wanted, render the scene to a muted `.mp4` and scroll-scrub it — Juan's approach — but start with the lighter CSS-3D hero.)

8. Accessibility Requirements (P0 — signature)

Target **WCAG 2.2 AA**. Must-haves:

- Valid `lang` attribute; per-route `<title>` .
 - One `<h1>` per page; logical heading order.
 - Landmarks: `<header>` `<nav>` `<main>` `<footer>` .
 - "Skip to content" link as the first focusable element.
 - Visible focus indicators everywhere (3px `--color-focus`); never `outline:none` without a replacement.
 - Full keyboard operability, no traps (nav, 3D scene, custom cursor included).
 - Text alternatives for meaningful media; decorative media `aria-hidden` .
 - Color is never the only signal.
 - Global, respected `prefers-reduced-motion: reduce` path: disables parallax, autoplay, scroll-jacking; offers equally-polished static fallbacks.
 - **Bonus credibility:** a visible footer "Accessibility" link with a short conformance statement + contact for issues.
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9. Performance Budget

- JS <= ~120KB gzipped on the hero route; Lenis/GSAP load only where needed.
- Images via `<picture>` AVIF/WebP with width/height set (avoid CLS).
- All video `preload="none"` , lazy-played in view, with posters.
- Fonts self-hosted, `font-display: swap` , subset.
- LCP element is real text (the `<h1>`), not the 3D scene.

10. Phased Build Plan / Milestones

Phase	Focus	Deliverable
0 — Foundations (wk 1)	Repo, Next+TS+Tailwind, tokens, base layout w/ landmarks, skip link, nav, bookend footer, reduced-motion plumbing, axe-core + Lighthouse CI	Deployed accessible skeleton
1 — Content + Work (wk 2)	Content model for projects + playground; sticky scroll-spy Work page (Challenge/Services/Role); real projects added	Highest-value page live
2 — Playground (wk 3)	CSS-columns masonry, lazy media, experiments + curation, date tags, focus polish	Playground live
3 — 3D Hero (wk 4)	CSS-3D diorama, mouse parallax, scroll ease-in, reduced-motion static fallback, <code><h1></code> /tagline as text	Hero live (built last; site usable without it)
4 — About + audit (wk 5)	About page (a11y philosophy), optional contact form, custom cursor (progressive), live clock; manual a11y pass (keyboard, VoiceOver/NVDA), perf tuning	Launch

11. References

- **William Le** — <https://williamle.design/> (3D entry) and /play (Playground). Built in Framer; CSS-3D transforms (matrix3d, rotateX ~20deg); 3-col masonry; respects reduced-motion; no `lang` /skip-link.
- **Juan Mora** — <https://juanmora.co/> (home) and /work. Built in Webflow; GSAP 3.15 + ScrollTrigger + SplitText + Lenis 1.3.17; sticky left scroll-spy; video-driven

cinematic hero (preload="none" , muted loop); shared bookend CTA + wordmark;
beige #FAF6EF light surface.

12. Open Questions

1. Monogram/wordmark: do you want a "JS" or "JMS" mark (like William's "W")?
2. Display typeface: Clash Display (proposed) or another?
3. Contact: email-only link, or a form (adds a serverless handler)?
4. Do you want individual deep case-study pages later, or keep links-to-live only?
5. Custom domain + analytics (privacy-friendly, e.g. Plausible)?